



**CS411W Team Bronze
Spring 2026
Prototype 1**

Spoiler Alert
‘No Surprises at the Table’

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Meet the Team!



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Project Lead/Database



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Backend



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Backend



Parker Mullenax
Frontend Developer



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Developer/Webmaster

The Problem

Food recalls happen all the time, however, there isn't an adequate reporting system overall. It is vital that everybody receives topical recall reports promptly, yet many do not. This is a major safety concern for consumers, as companies don't take proper precautions.

Despite official resources being a thing, their own notifications are restricted to email, not an optimal form in the age of the smartphone. Existing apps also fall short, not being personalized or appealing enough to the average user to be noteworthy.



Solution Statement

Spoiler Alert: An innovative web & mobile application that delivers personalized, immediate, and trustworthy food recall alerts.



- ★ 🕒 **Speed & Urgency** → Instant notifications (push, email).
- ★ 📄 **Clarity** → Simple, consumer-friendly language for critical information.
- ★ 🎯 **Personalization** → Allergy filters, barcode/receipt scanning, custom channels.
- ★ 🔒 **Trust** → Direct FDA/USDA data in one reliable platform.

Result: Families, caregivers, and vulnerable individuals stay safe — no surprises at the table.

Changes to Scope

- We have switched to SQLite for our database, and have implemented Prisma as our ORM.
- Will Add Login-Based Recall Notifications - Users will receive a pop-up notification upon login if new relevant recalls are available.
- The Home Page is being restructured into the Landing Page and the Recalls Page will take over its functionality.
- We are lowering the priority of the employee dashboard and possible removing it entirely to focus on other user-based features.
- The OpenFoodFacts API is being added alongside BarcodeLookup for more consistent UPC finding.

Category	Feature	Description	Progress
Account Management	Account Registration with Email	Users should be able to create new accounts on the signup page. This will require an email and a secure password.	Finished
	Account Login	Users should be able to access their created account through the login page. This will require their email and the password they used to signup.	Finished
	Guest Login	New users without accounts can explore the application and see if its the right fit for them.	Finished
Notifications	Email Alerts	Users should receive emails detailing any recalls that their settings allow them to receive. By default, this will email the user about all of them.	Finished
	Alert Filtering	Users should be able to have filters set up to block the receiving of alerts that they do not want.	In Progress
	Pop-Up Alerts	Users should receive a pop-up notification upon logging in if there are new recalls that match their settings since their last login.	Finished

Feature Table 1



Category	Feature	Reasoning	Progress
Item Tracking	Inventory Viewing	Users should be able to view products that they have “scanned” (registered) with the application.	Finished
	Inventory Item Adding	Users should be able to add an inventory to their inventory by entering a UPC into a form.	Finished
	Inventory Item Removal	Users should be able to remove a product from their inventory if they do not want to be alerted for recalls about that product anymore.	Finished
Searching	Inventory Product Search	Users should be able to search their inventory by product name or brand.	Finished
	Keyword Recall Search	Users should be able to search the homepage recall list by recall reason or product.	Not Started

Feature Table 2

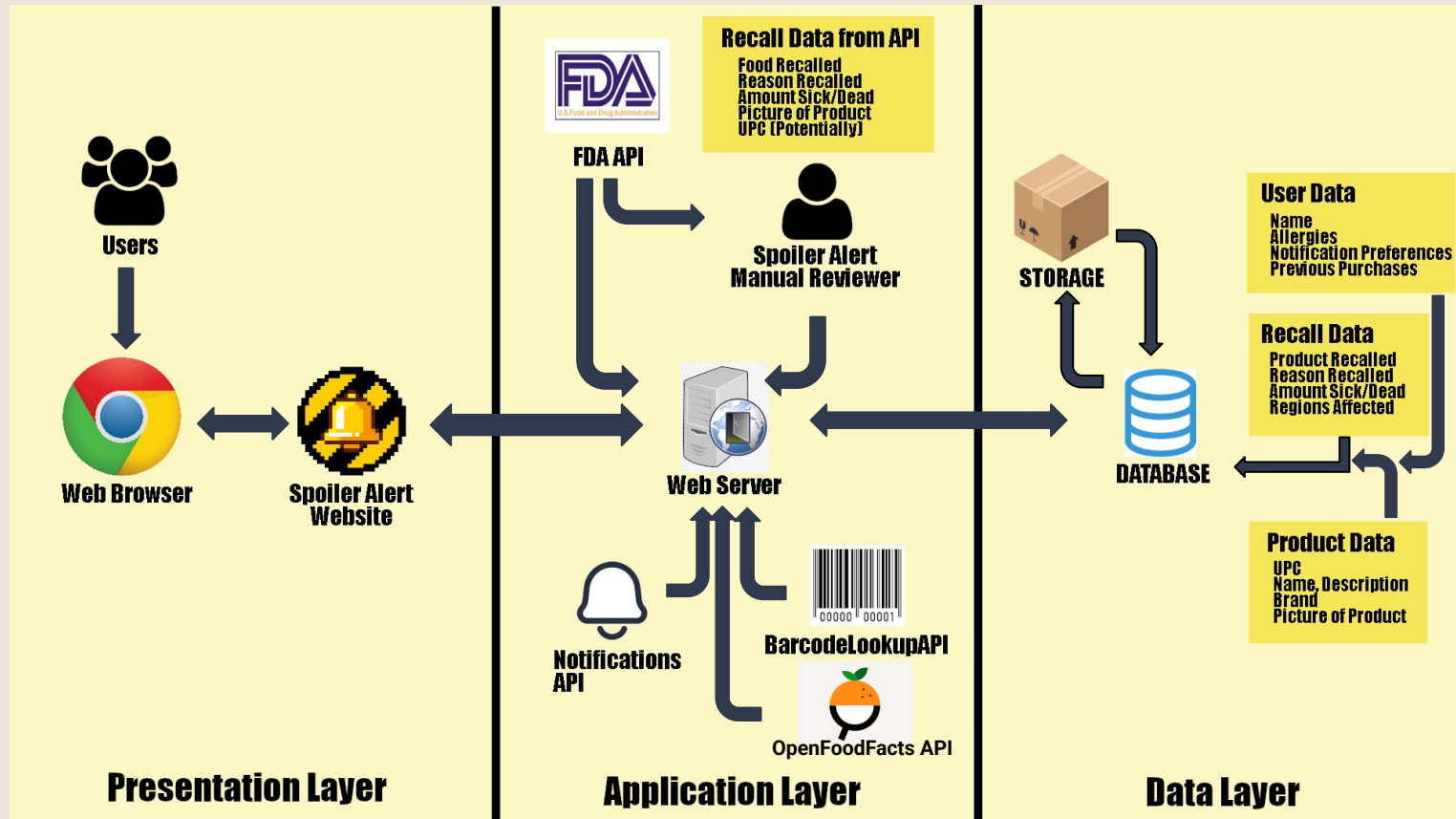


Category	Feature	Reasoning	Progress
Recalls	Recall Timeline	Users should get a list of recent recalls on their recalls page.	In Progress
	Filter Recalls	Users should be able to filter the recalls in regards to their filtering settings. This can take the form of filtering out mixed in allergens that do not matter to you or filtering out product types entirely.	Not Started
	Recall Details	Users should be able to click on a recall to view more information about the recall, including a link to official sources.	Finished
Recall Reviewer Dashboard	UPC Not Found Notification	Employees should be notified of a product or recall that needs a UPC but does not have one yet.	Not Started
	UPC Linking	Employees should be able to add a UPC to a product or recall that does not have one yet.	Not Started
	Manual Recall Editing	Employees should be able to adjust recall information in the event our algorithms do not get all the information they need.	Not Started

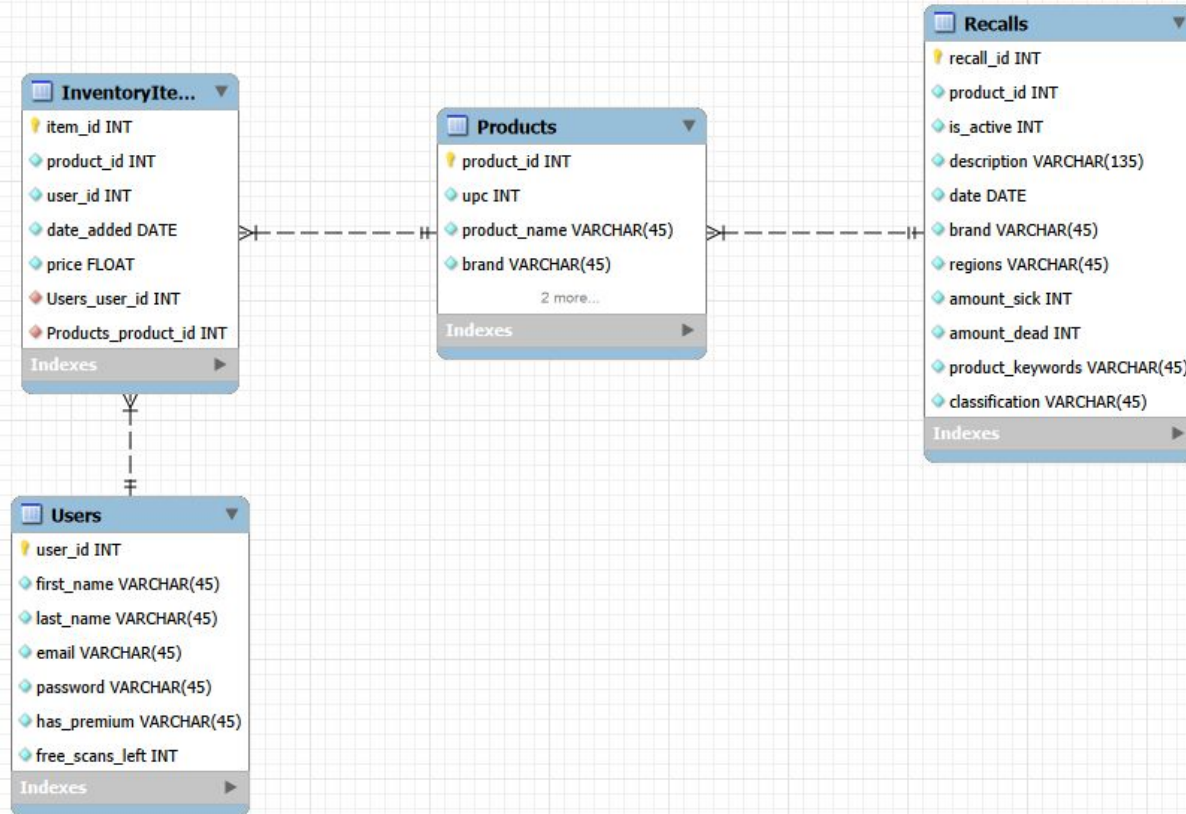
Feature Table 3



Major Functional Components



Database Schema

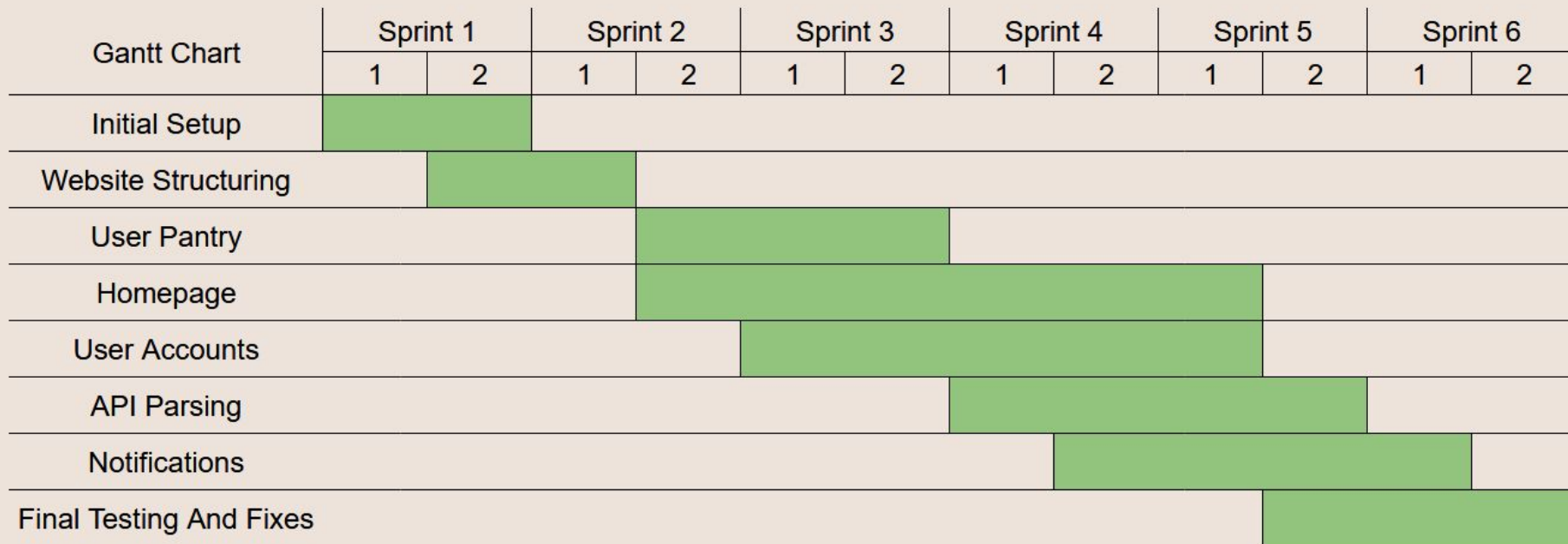


Development Tools



Communication	Discord
Version Control	Git / GitHub
IDE	VSCode
Collaboration	ClickUp
Front End	Bootstrap / EJS
Back End	NodeJS
Database	Prisma / SQLite

Sprint Breakdown



Sprint Breakdown (2 Week Increments)

Sprint 1

- ❑ Setup code repositories
- ❑ Prepare development tools and frameworks
- ❑ Setup initial database, referencing initial schema
- ❑ Prepare initial pages for the website

Sprint 2

- ❑ Load and display recall data into homepage feed.
- ❑ Load and display product data into pantry.
- ❑ Load different product data based on user id.

Sprint 3

- ❑ Add new items to pantry inventory
- ❑ Remove items from pantry inventory
- ❑ Begin work on email notification system.
- ❑ Save new user data for account creation and login capability

Sprint Breakdown (2 Week Increments)

Sprint 4

- ❑ Begin parsing openFDA JSON data.
- ❑ Automatically load parsed recall data into database.
- ❑ Notification customization based on "User Recall Filter" table
- ❑ Finish formatting on Pantry page

Sprint 5

- ❑ Finish user accounts
- ❑ Complete email notification pipeline
- ❑ Finish formatting recall listings with detailed information
- ❑ New recall handling

Sprint 6

- ❑ Final UI polish and cleanup
- ❑ Optimize performance
- ❑ Final testing and documentation

User Risks



Risk Factors:

R1: Users may suffer from alert fatigue, from recalls concerning products not relevant to the user.

R2: Users may forget to turn push notifications on, causing a delay in information.

R3: Regional differences may cause misinformation.

R4: Users may not want to share their personal medical information with the app.

Mitigations:

M1: Spoiler Alert customizes reports, so the user only gets notified of the reports that are relevant to them.

M2: Spoiler Alert will alert users that their notifications are off whenever they open the app.

M3: Spoiler Alert will ensure that the regions affected will be one of the first things users see when looking at a recall.

M4: Spoiler Alert will ensure users that the app will encrypt and responsibly store the data. Users may also continue to use the app without entering this data at all.

		Likelihood				
		Very Low	Low	Medium	High	Very High
Impact	Severe					
	High			R1	R4	
	Medium		R3	R2		
	Low		M1 M2	M4		
	Very Low	M3				

Technical Risks



Risk Factors:

R1: Certain stores may have unique product code formats on their receipts that the app will have trouble looking up.

R2: Low user camera quality can result in unreadable scanning

R3: APIs may have downtime or contain broken data, slowing down or stopping Spoiler Alert's ability to report.

R4: Most recall APIs (FDA, USDA, CPSC) do not include UPCs in their metadata, making it difficult for Spoiler Alert to match scanned product barcodes to active recalls.

Mitigations:

M1: Spoiler Alert will work with the bigger stores to make those product codes searchable for recall logging.

M2: In the case camera quality makes the receipt unreadable, the user can manually input the product code.

M3: If APIs for any reason are unreadable, the servers will attempt to read from the API a few more times. If this still fails, sources will be manually reviewed by staff until the API works once more.

M4: Spoiler Alert will maintain its own UPC database from manufacturer sites, official recall web pages, and product catalogs when the recall APIs don't include it.

		Likelihood				
		Very Low	Low	Medium	High	Very High
Impact	Severe			R3		
	High			R1	R4	
	Medium	M3				
	Low		R2 M1	M4		
	Very Low	M2				

Legal/Security Risks



Risk Factors:

- R1:** Personal information, such as medical and payment information, can be subject to data leaks.
- R2:** Companies may try to litigate Spoiler Alert for reports that damage their image.
- R3:** The app may collect sensitive personal and medical information without consent.
- R4:** User purchases a product that gets recalled later on. SpoilerAlert may not connect the purchase to the recall and will fail to inform the user.

Mitigations:

- M1:** SpoilerAlert uses up-to-date security protocol to ensure against data attacks.
- M2:** Companies should be held accountable for harm regardless of litigation.
- M3:** App informs and gets users consent with terms of service and privacy policy agreement upon account creation.
- M4:** Popup disclaimer and user agreement to warn about possible missed recalls and limit SpoilerAlert's liability.

Impact

		Likelihood				
		Very Low	Low	Medium	High	Very High
Impact	Severe				R4	
	High	R1			R3	
	Medium			M4		
	Low	M1				R2
	Very Low		M2 M3			

Any Questions?



Next Steps: Parker Davis

FDA Recall API - Will make sure that SpoilerAlert is updated automatically with the FDA API, and that the data is understood by the server.

Pantry Searching Backend - Will implement the searching algorithms so that the user can search and filter their pantry.

Next Steps: Mark Stoegbauer

Ability To Filter Recalls Page - Allow users to filter their Recalls page based on their settings and preferences.

Ability to Filter Pantry Page - Allow users to filter their Pantry page to look for particular items they have saved/purchased.

Add Personalized Recall Section - Display a personalized section on the recalls page for logged-in users, showing recalls that match their filtering preferences.

Next Steps: Jackie Simmons

Set up settings page- Edit notifications preference page

Create new page for recalls and move from homepage- move homepage functionality for recalls over to recall page

Next Steps: Parker Mullenax

Refactor Homepage and Recalls Page - Move recall feed from the homepage to the dedicated recalls page. A new landing page will be created to serve as the main entry point for users.

Implement Searching UI - Add the ability for the user to search for recalls and pantry items using a search input.

Implement Filtering UI - Add the ability for users to set recall filters based on allergens or dietary restrictions

Add Login Notification Pop-Up - Display a pop-up notification when a user logs in if new recalls matching their filtering preferences have been issued since their last login.

Next Steps: Shan Ibanez



Searching:

API: Allow user to search product by name, brand, and UPC.

Notifications:

Email Alerts: Send all new recalls by default.

Alert Filtering: User can choose what types of recall they do not want to receive.

Pop-Up Alerts: Only send emails that user wants to receive.

Next Steps: Joshua adkins

Update settings page - Add customized notification settings for users

Recall Filtering - Filter by allergen, region, food category

Recall stats - Recalls page lists all recent recalls depending on time frame

Recall search - Search through recalls?

Recall toggle - Flip between pantry and open recalls

Landing page?

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